Volumes of Revolution

Disks, washers and cylindrical shells

Calculus II handout · Yitian Zhang Formulas and rules Worked examples Notes and cautions

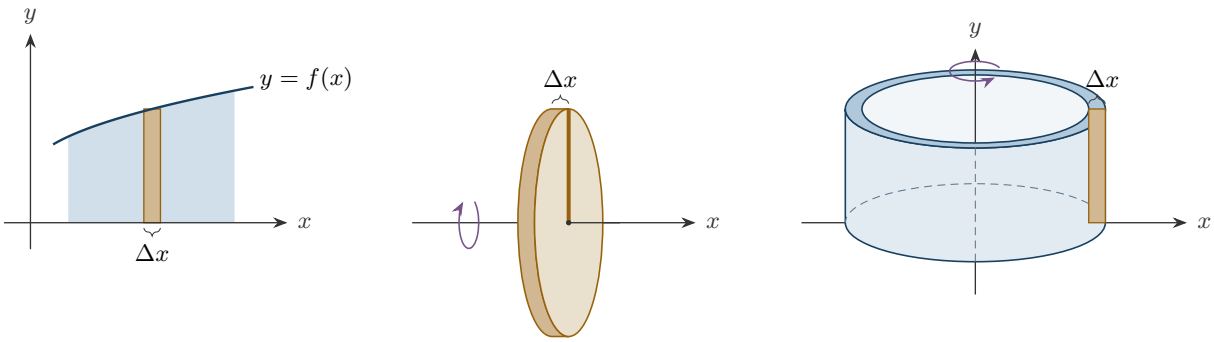
1 One idea, three shapes

A *solid of revolution* is the solid swept out when a flat region is rotated once around a line, called the *axis of rotation*. Every method in this handout finds its volume in the same three steps.

1. Cut the region into thin strips.
2. Rotate one strip around the axis. It sweeps out a thin solid whose volume is easy to write down.
3. Add up all the thin solids with an integral.

The only thing that changes from one method to the next is the shape of the thin solid, and that depends on how the strip sits relative to the axis.

Strip, compared with the axis	It sweeps out	Volume of that piece
perpendicular, touching it	a disk	$\pi\,(\text{radius})^2 \cdot \text{thickness}$
perpendicular, not touching it	a washer (a disk with a hole)	$\pi\,(R^2 - r^2) \cdot \text{thickness}$
parallel	a cylindrical shell	$2\pi\,(\text{radius})(\text{height}) \cdot \text{thickness}$



(a) one vertical strip (b) about the x-axis: a disk (c) about the y-axis: a shell

Figure 1. The same vertical strip gives a disk when it is rotated about the x-axis (it is perpendicular to that axis) and a cylindrical shell when it is rotated about the y-axis (it is parallel to that axis).

The thickness of the strip is the variable of integration

A vertical strip has thickness $\mathrm{d}x$, so the radius, the height and the limits must all be written in terms of x. A horizontal strip has thickness $\mathrm{d}y$, so everything is written in terms of y.

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2 The disk method

Take the region under $y = f(x)$ and above the x-axis, for $a \le x \le b$, and rotate it about the x-axis. A vertical strip at position x is perpendicular to the axis and reaches it. Rotated, it sweeps out a flat disk, like a coin, of radius $f(x)$ and thickness Δx. Its volume is the area of its circular face times its thickness: $\pi\,[f(x)]^2\,\Delta x$.

Disk method

About the x-axis, with vertical strips of height $f(x)$ for $a \le x \le b$:

$$V = \int_a^b \pi\,[f(x)]^2\,\mathrm{d}x.$$

About the y-axis, with horizontal strips of length $g(y)$ for $c \le y \le d$:

$$V = \int_c^d \pi\,[g(y)]^2\,\mathrm{d}y.$$

In both cases the radius is the length of the strip, and you integrate *along* the axis.

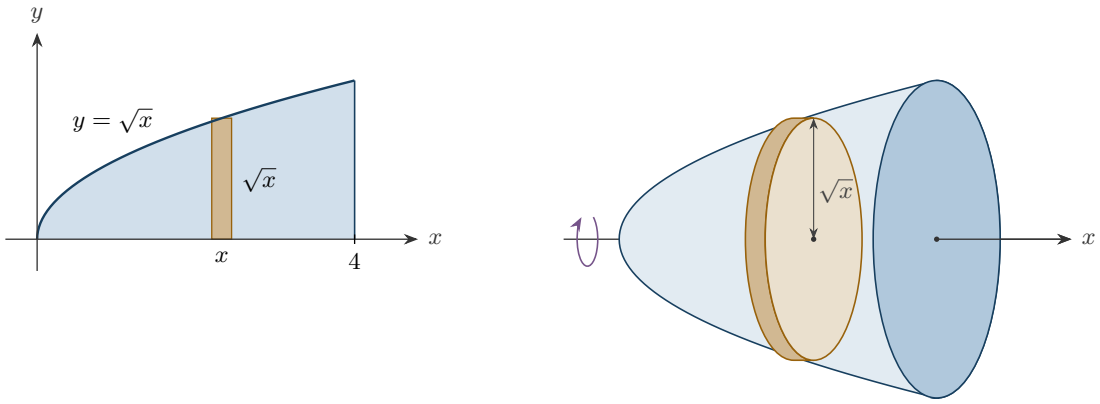


Figure 2. The region under $y = \sqrt{x}$ for $0 \le x \le 4$, and the solid it sweeps out about the x-axis. The strip at x becomes a disk of radius $\sqrt{x}$.

Example 1 ▪ a first volume disks

Find the volume of the solid obtained by rotating the region under $y = \sqrt{x}$, $0 \le x \le 4$, about the x-axis (Figure 2).

Solution. A vertical strip at x is perpendicular to the x-axis and touches it, so it sweeps out a disk of radius $\sqrt{x}$.

$$V = \int_0^4 \pi(\sqrt{x})^2\,\mathrm{d}x = \pi\int_0^4 x\,\mathrm{d}x = \pi\Big[\frac{x^2}{2}\Big]_0^4 = \boxed{8\pi}.$$

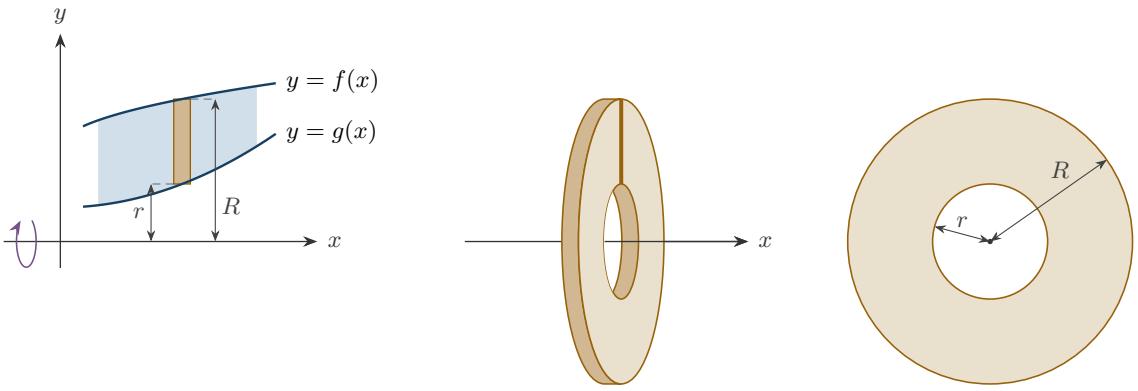
3 The washer method

If the strip does not reach the axis, the solid has a hole in it, and the strip sweeps out a *washer*: a disk of radius R with a smaller disk of radius r removed from its centre.

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- The *outer radius* R is the distance from the axis to the far end of the strip.
- The *inner radius* r is the distance from the axis to the near end of the strip.

The face of the washer has area $\pi R^2 - \pi r^2$, so one washer has volume $\pi\,(R^2 - r^2)\,\Delta x$.



(a) the strip stops short of the axis

(b) it sweeps out a washer

(c) its face has area $\pi R^2 - \pi r^2$

Figure 3. The washer method. R and r are both measured from the axis of rotation.

Washer method

$$V = \int_a^b \pi\,(R^2 - r^2)\,\mathrm{d}x \qquad \text{(or the same with } \mathrm{d}y\text{, for horizontal strips).}$$

For the region between $y = f(x)$ on top and $y = g(x)$ below, with $f \geq g \geq 0$, rotated about the x-axis: $R = f(x)$ and $r = g(x)$. A disk is the special case $r = 0$.

Caution: $R^2 - r^2$ is not $(R - r)^2$

You subtract two areas, $\pi R^2 - \pi r^2$. The length of the strip, $R - r$, is never squared.

Rotating about another line. Nothing changes except that R and r are distances to the new axis, not to the x-axis. About the horizontal line $y = k$, a point at height y is at distance $|y - k|$.

Example 2 ▪ a solid with a hole washers

The region between $y = x$ and $y = x^2$ is rotated about the x-axis. Find the volume.

Solution. The curves meet at $x = 0$ and $x = 1$, and $x \geq x^2$ in between. A vertical strip at x runs from $y = x^2$ up to $y = x$ and does not touch the axis, so $R = x$ and $r = x^2$.

$$V = \int_0^1 \pi(x^2 - x^4)\,\mathrm{d}x = \pi\Big[\frac{x^3}{3} - \frac{x^5}{5}\Big]_0^1 = \pi\Big(\frac{1}{3} - \frac{1}{5}\Big) = \boxed{\frac{2\pi}{15}}.$$

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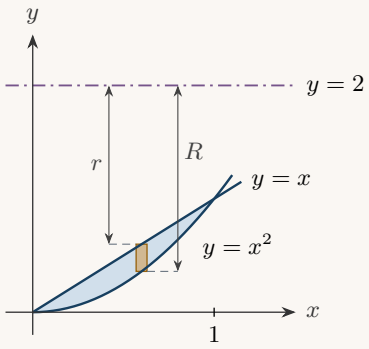
Example 3 ▪ the same region about $y = 2$ washers

The region between $y = x$ and $y = x^2$ is rotated about the line $y = 2$. Find the volume.

Solution. The strip is the same as in Example 2, but now the axis is above the region.

The far end of the strip is on the parabola and the near end is on the line, so

$$R = 2 - x^2, \qquad r = 2 - x.$$



$$\begin{aligned} V &= \int_0^1 \pi\Big[(2-x^2)^2 - (2-x)^2\Big]\,dx \\ &= \pi\int_0^1 (x^4 - 5x^2 + 4x)\,dx \\ &= \pi\Big[\frac{x^5}{5} - \frac{5x^3}{3} + 2x^2\Big]_0^1 = \pi\Big(\frac{1}{5} - \frac{5}{3} + 2\Big) = \boxed{\frac{8\pi}{15}}. \end{aligned}$$

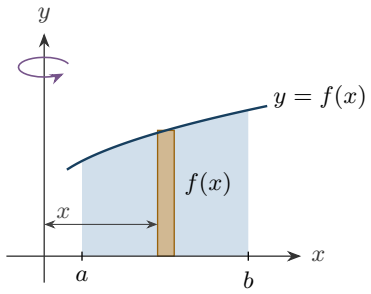
4 The shell method

4.1 What a shell is

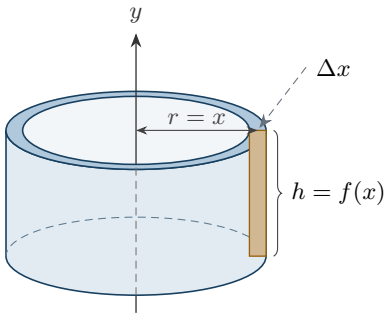
Now rotate the region under $y = f(x)$, $a \le x \le b$ with $a \ge 0$, about the y-axis. A vertical strip at position x is *parallel* to the axis. As it goes around, it sweeps out a thin hollow cylinder, like a tin can with no top and no bottom. This is a *cylindrical shell*, with

$$\text{radius } r = x, \qquad \text{height } h = f(x), \qquad \text{wall thickness } \Delta x.$$

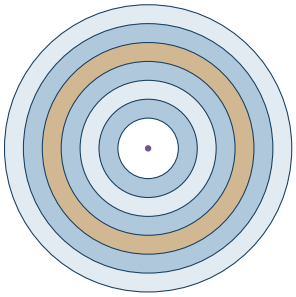
The shells sit inside one another like the rings of a tree trunk, and together they fill the solid.



(a) the strip is parallel to the axis



(b) it sweeps out a shell



(c) from above: nested shells

Figure 4. The shell method about the y-axis. The radius of a shell is the distance from the axis to the strip, and its height is the length of the strip.

4.2 Where $2\pi rh$ comes from

Take one shell of radius r, height h and wall thickness Δr. Cut it straight down one side, from top to bottom, and unroll it flat, the way a paper tube opens into a sheet of paper (Figure 5). What you get

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is a thin rectangular box, with

$$\text{length} = \text{circumference of the shell} = 2\pi r, \qquad \text{height} = h, \qquad \text{thickness} = \Delta r,$$
$$\text{volume of one shell} \;\approx\; \text{length} \times \text{height} \times \text{thickness} \;=\; 2\pi r \cdot h \cdot \Delta r.$$

So $2\pi rh$ is the area of the unrolled sheet (the side area of a cylinder), and the volume of the shell is that area times the thickness of the sheet.

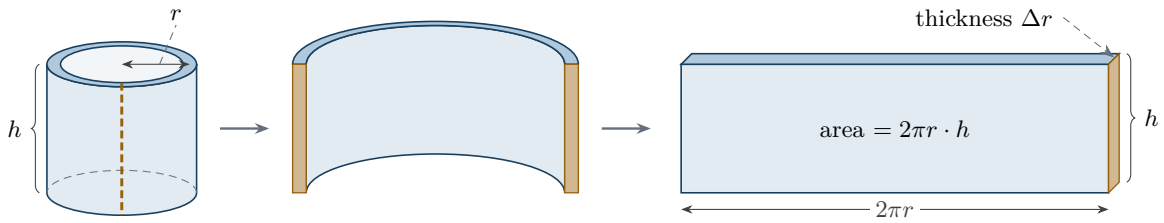


Figure 5. Unrolling a cylindrical shell. The picture is to scale: the box in (c) is exactly as long as the circumference of the shell in (a).

4.3 Which radius? The middle one

A wall with real thickness has an inner radius r_1 and an outer radius r_2. Its inner face has length $2\pi r_1$ and its outer face has length $2\pi r_2$, so the unrolled wall is not exactly a rectangle: seen from above it is closer to a trapezoid (Figure 6). A trapezoid has the same area as the rectangle built on its middle line. So measure the circumference at the *middle* of the wall, at the radius $\bar{r} = \frac{1}{2}(r_1 + r_2)$, and nothing is lost. The algebra confirms that this is exact:

$$V_{\text{shell}} = \pi r_2^2 h - \pi r_1^2 h = \pi\,(r_2 + r_1)(r_2 - r_1)\,h = 2\pi \cdot \underbrace{\frac{r_1 + r_2}{2}}_{\bar{r}} \cdot h \cdot \underbrace{(r_2 - r_1)}_{\Delta r} = 2\pi\,\bar{r}\,h\,\Delta r.$$

In the integral the walls become infinitely thin: r_1, $\bar{r}$ and r_2 all become the same x, and Δr becomes $\mathrm{d}x$.

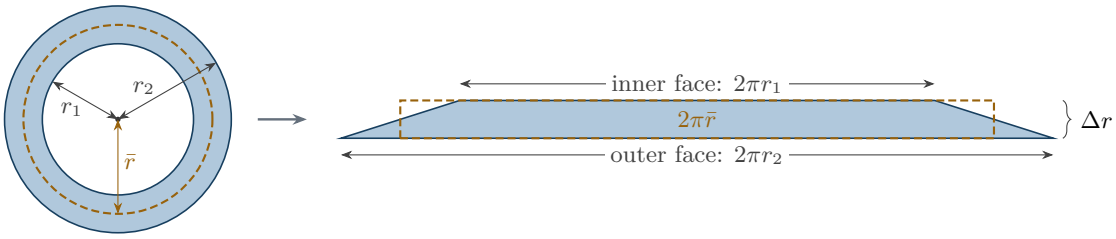


Figure 6. The wall of a shell seen from above, and unrolled. The dashed rectangle on the middle line has exactly the same area as the ring: $\pi r_2^2 - \pi r_1^2 = 2\pi\bar{r}\,\Delta r$.

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4.4 The formula

Shell method

$$V = \int_a^b 2\pi\,(\text{radius})\,(\text{height})\,dx = \int_a^b 2\pi\, x\, f(x)\,dx \qquad (\text{about the } y\text{-axis},\ 0 \le a \le b).$$

The radius is the distance from the axis to the strip. The height is the length of the strip. You integrate *across* the axis, from the smallest radius to the largest.

The same formula covers every other case, as long as the radius and the height are read off the picture.

- **Region between two curves:** the height is top minus bottom, $f(x) - g(x)$.
- **About the vertical line $x = c$:** the radius is the distance to that line, $|x - c|$. That is $c - x$ if the region is to the left of the line and $x - c$ if it is to the right.
- **About a horizontal axis** (the x-axis, or $y = k$): the strips parallel to the axis are horizontal. The radius is $|y - k|$, the height is the length of the strip, right minus left, and the integral is in y.

Example 4 ▪ one solid, both methods shells, then washers

The region between $y = x$ and $y = x^2$ is rotated about the y-axis. Find the volume.

Solution. **Shells.** A vertical strip at x is parallel to the y-axis. Its radius is x and its height is top minus bottom, $x - x^2$.

$$V = \int_0^1 2\pi\, x\,(x - x^2)\,dx = 2\pi \int_0^1 (x^2 - x^3)\,dx = 2\pi\Big(\frac{1}{3} - \frac{1}{4}\Big) = \boxed{\frac{\pi}{6}}.$$

Check with washers. A horizontal strip at height y is perpendicular to the y-axis. It runs from the line, where $x = y$, to the parabola, where $x = \sqrt{y}$. So $R = \sqrt{y}$, $r = y$, and

$$V = \int_0^1 \pi(y - y^2)\,dy = \pi\Big(\frac{1}{2} - \frac{1}{3}\Big) = \frac{\pi}{6}.$$

Both methods always give the same volume. Here both are easy, because each curve is easy to solve for x.

5 Which method?

In principle both methods work for every solid. Choose the one whose integral is easier, and that is decided by the direction of the strip.

Axis of rotation	**Disks or washers** (strip $\perp$ axis)	**Shells** (strip $\parallel$ axis)
horizontal (x-axis, $y = k$)	vertical strips, integrate in x	horizontal strips, integrate in y
vertical (y-axis, $x = c$)	horizontal strips, integrate in y	vertical strips, integrate in x

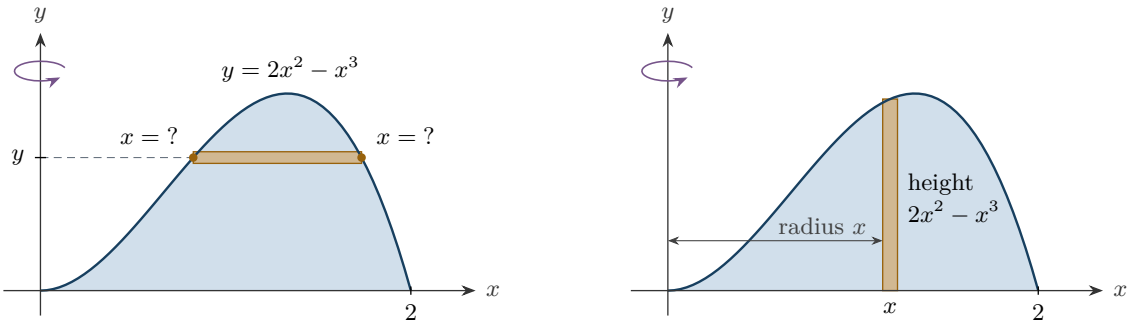
When the curves are given as $y = f(x)$, vertical strips are the natural ones, because their ends are already known: top $f(x)$, bottom $g(x)$. Read the table with that in mind. About a horizontal axis, vertical strips mean washers. About a vertical axis, vertical strips mean shells.

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Rule: a curve that is not injective, rotated about a vertical axis ⇒ shells

Suppose the region is bounded by a curve $y = f(x)$ where f is *not injective* (not one-to-one) on the interval. Its graph turns around, for example it goes up and comes back down, so some horizontal line meets it more than once. If the axis of rotation is *vertical*, use the shell method.

Why. Washers about a vertical axis need horizontal strips. A horizontal strip at height y starts and ends on the curve, so its two ends are two different values of x with the same y, and you would have to solve $y = f(x)$ for x once for each branch of the curve. A function that is not injective has no single inverse. Often there is no usable formula at all: try solving $y = 2x^2 - x^3$ for x (Figure 7). When there is one, each branch needs its own formula, and the integral gets harder (Example 6). A vertical strip avoids all of this. It is parallel to the axis, its height is simply $f(x)$ and its radius is simply x.



(a) washers: a horizontal strip has both ends on the same curve, and neither end has a formula

(b) shells: a vertical strip has radius x and height $f(x)$, with nothing to solve

Figure 7. The region under $y = 2x^2 - x^3$, rotated about the y-axis. The curve fails the horizontal line test, so horizontal strips are the wrong choice and shells are the right one.

Two remarks on the rule

- It has a mirror image. If the curve is $x = g(y)$ with g not injective and the axis is *horizontal*, use shells with horizontal strips.
- It is a preference, not a law. Washers still give the right answer whenever you can find both branches (Example 6). And when f is injective, both methods are usually fine (Example 4).

Example 5 ▪ no inverse available shells

The region bounded by $y = 2x^2 - x^3$ and the x-axis is rotated about the y-axis. Find the volume.

Solution. The curve meets the x-axis at $x = 0$ and $x = 2$, rises to a maximum at $x = \frac{4}{3}$ and comes back down, so it is not injective on $[0, 2]$. The axis is vertical: use shells. A vertical strip at x has radius x and height $2x^2 - x^3$ (Figure 7b).

$$V = \int_0^2 2\pi\, x\,(2x^2 - x^3)\,dx = 2\pi \int_0^2 (2x^3 - x^4)\,dx = 2\pi\Big[\frac{x^4}{2} - \frac{x^5}{5}\Big]_0^2 = 2\pi\Big(8 - \frac{32}{5}\Big) = \boxed{\frac{16\pi}{5}}.$$

With washers you would need the two solutions x of the cubic equation $x^3 - 2x^2 + y = 0$ as functions of y.

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Example 6 ▪ what the other method costs shells, then washers

The region under $y = \sin x$, $0 \le x \le \pi$, is rotated about the y-axis. Find the volume.

Solution. $\sin x$ is not injective on $[0, \pi]$ and the axis is vertical.

Shells. Radius x, height $\sin x$. Integrating by parts,

$$V = \int_0^{\pi} 2\pi\, x \sin x \, dx = 2\pi \Big[-x\cos x + \sin x\Big]_0^{\pi} = 2\pi \cdot \pi = \boxed{2\pi^2}.$$

Washers. A horizontal strip at height y runs from the left branch $x = \arcsin y$ to the right branch $x = \pi - \arcsin y$, so $r = \arcsin y$ and $R = \pi - \arcsin y$.

$$V = \int_0^1 \pi\Big[(\pi - \arcsin y)^2 - (\arcsin y)^2\Big]\, dy = \pi \int_0^1 (\pi^2 - 2\pi \arcsin y)\, dy$$
$$= \pi^3 - 2\pi^2 \Big[y \arcsin y + \sqrt{1-y^2}\Big]_0^1 = \pi^3 - 2\pi^2\Big(\frac{\pi}{2} - 1\Big) = 2\pi^2.$$

The answer is the same, at the cost of two inverse branches and the integral of arcsin.

A routine that always works

1. Sketch the region and draw the axis of rotation.
2. Decide the direction of the strip: which direction gives a single formula for each end?
3. Compare the strip with the axis. Perpendicular means disks or washers. Parallel means shells.
4. Read each radius as a distance to the axis, and the height as the length of the strip.
5. Find the limits: the values of x (or of y) between which the strips run.
6. Integrate.

6 Two more examples

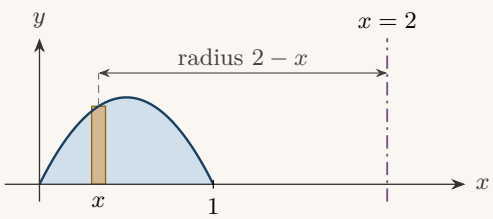
Example 7 ▪ an axis that is not a coordinate axis shells

The region bounded by $y = x - x^2$ and the x-axis is rotated about the line $x = 2$. Find the volume.

Solution. The curve is a parabola that is not injective on $[0, 1]$, and the axis is vertical: use shells.

The region lies to the left of the axis, so a vertical strip at x has

radius $2 - x$, height $x - x^2$.



$$V = \int_0^1 2\pi\,(2-x)(x-x^2)\, dx = 2\pi \int_0^1 (x^3 - 3x^2 + 2x)\, dx = 2\pi\Big(\frac{1}{4} - 1 + 1\Big) = \boxed{\frac{\pi}{2}}.$$

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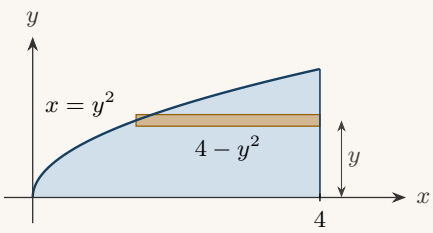
Example 8 ▪ shells about the x-axis shells

Find the volume of Example 1 again, this time with shells: the region under $y = \sqrt{x}$, $0 \le x \le 4$, rotated about the x-axis.

Solution. Strips parallel to the x-axis are horizontal.

A horizontal strip at height y, for $0 \le y \le 2$, runs from the curve, where $x = y^2$, to the line $x = 4$. Its radius is y and its length is $4 - y^2$. The answer is the same as with disks:

$$V = \int_0^2 2\pi\, y\,(4 - y^2)\,\mathrm{d}y = 2\pi\Big[2y^2 - \frac{y^4}{4}\Big]_0^2 = \boxed{8\pi}.$$



7 Summary

The three formulas

Disk	strip perpendicular to the axis, touching it	$V = \int \pi\,(\text{radius})^2\,\mathrm{d}x$
Washer	strip perpendicular to the axis, not touching it	$V = \int \pi\,(R^2 - r^2)\,\mathrm{d}x$
Shell	strip parallel to the axis	$V = \int 2\pi\,(\text{radius})(\text{height})\,\mathrm{d}x$

These are written for vertical strips. For horizontal strips, replace $\mathrm{d}x$ by $\mathrm{d}y$ and write everything in terms of y. Disks and washers are integrated along the axis. Shells are integrated across it, from the smallest radius to the largest. A curve $y = f(x)$ that is not injective, rotated about a vertical axis, calls for shells.

8 Practice

Set up each integral from a sketch, then evaluate it.

1. The region under $y = x^2$, $0 \le x \le 2$, about the x-axis.
2. The same region, about the y-axis. Do it with shells, then with washers.
3. The region between $y = x^2$ and $y = \sqrt{x}$, about the x-axis.
4. The region between $y = 4x - x^2$ and the x-axis, about the y-axis.
5. The region under $y = \sin(x^2)$, $0 \le x \le \sqrt{\pi}$, about the y-axis.
6. The region under $y = e^{-x^2}$, $0 \le x \le 1$, about the y-axis.
7. The region between $y = x$ and $y = x^2$, about the line $x = -1$.
8. The region between $y = x^2$ and the line $y = 4$, about the line $y = 4$.
9. A sphere of radius a, in two ways. With disks, rotate the region under $y = \sqrt{a^2 - x^2}$, $-a \le x \le a$, about the x-axis. With shells, rotate the half-disk $x^2 + y^2 \le a^2$, $x \ge 0$, about the y-axis.

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Answers

1. Disks: $\displaystyle\int_0^2 \pi x^4 \, dx = \frac{32\pi}{5}$.

2. Shells: $\displaystyle\int_0^2 2\pi x \cdot x^2 \, dx = 8\pi$.

Washers: $\displaystyle\int_0^4 \pi \, (4 - y) \, dy = 8\pi$.

3. Washers: $\displaystyle\int_0^1 \pi \, (x - x^4) \, dx = \frac{3\pi}{10}$.

4. Shells: $\displaystyle\int_0^4 2\pi x \, (4x - x^2) \, dx = \frac{128\pi}{3}$.

5. Shells: $\displaystyle\int_0^{\sqrt{\pi}} 2\pi x \sin(x^2) \, dx = 2\pi$.

6. Shells: $\displaystyle\int_0^1 2\pi x \, e^{-x^2} \, dx = \pi\Big(1 - \frac{1}{e}\Big)$.

7. Shells, radius $x + 1$:
$\displaystyle\int_0^1 2\pi \, (x + 1)(x - x^2) \, dx = \frac{\pi}{2}$.

8. Disks, radius $4 - x^2$:
$\displaystyle\int_{-2}^2 \pi \, (4 - x^2)^2 \, dx = \frac{512\pi}{15}$.

9. Disks: $\displaystyle\int_{-a}^a \pi \, (a^2 - x^2) \, dx = \frac{4\pi a^3}{3}$.

Shells: $\displaystyle\int_0^a 2\pi x \cdot 2\sqrt{a^2 - x^2} \, dx = \frac{4\pi a^3}{3}$.

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